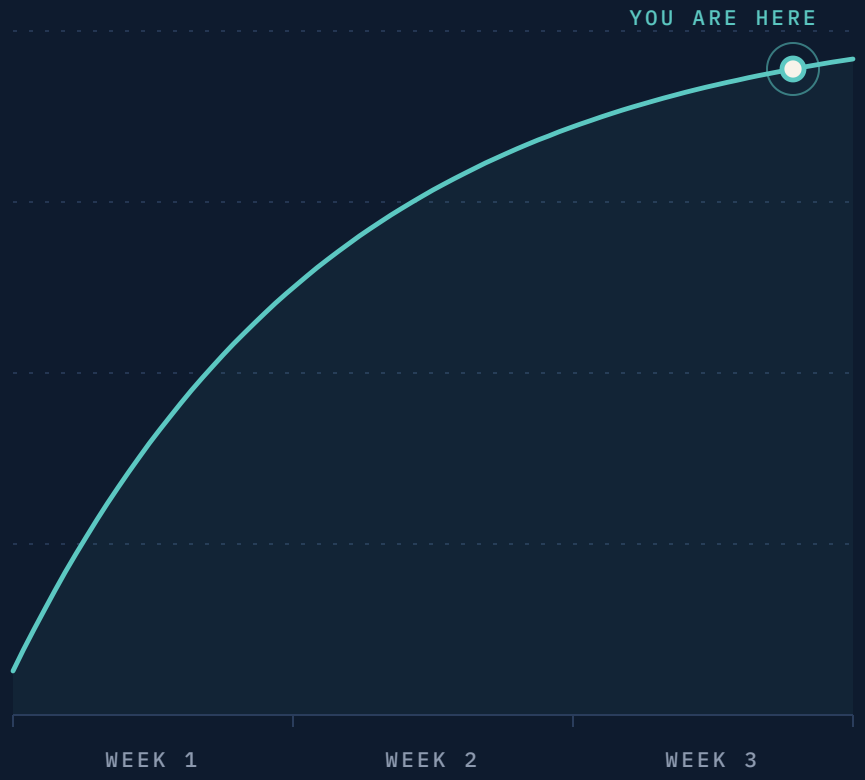
WEEK 3 · DAY 4 · OVERVIEW

# Portfolio Aggregation and Scenarios




Created by **Jeff Lenamon**

ABOUT 50 MIN WITH EXERCISES / 8 SECTIONS / 5 EXERCISES

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

# The Desk Lead's Note

The September book, 200 positions. I want the risk report's top lines from your own tested functions: value, duration, spread, DTS, convexity, DV01. Then the what-ifs, repriced in full: every yield -100, -25, +25 and +100 bp; every spread +25 bp and +10 percent. How far off is the duration-and-convexity shortcut? Which sectors carry each loss? Send me the grid as a workbook.

ABOUT 50 MIN WITH EXERCISES / 8 SECTIONS / 5 EXERCISES

# Your Route Through Day 4

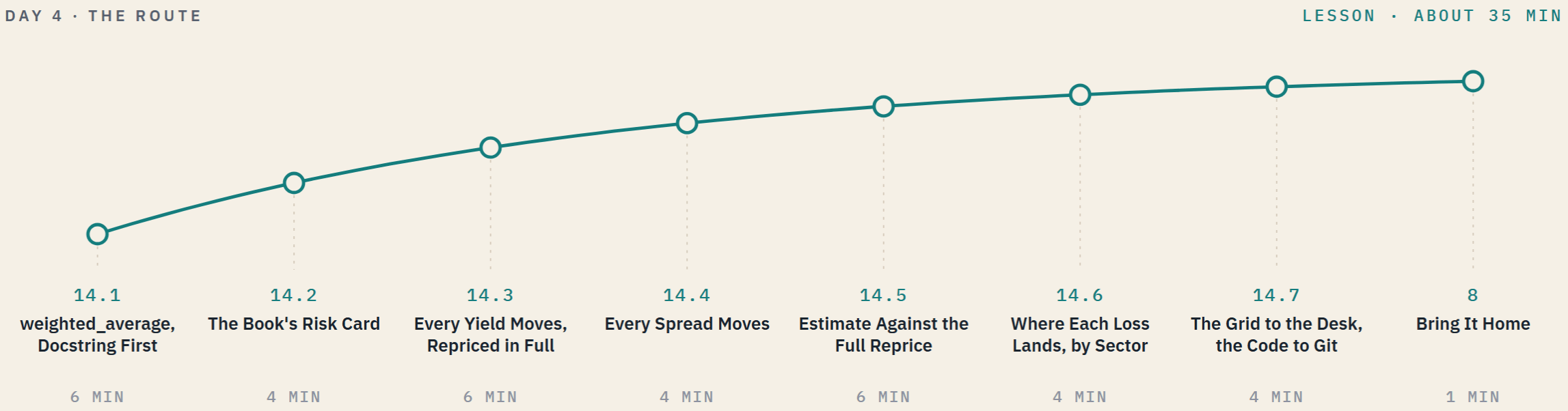



FIGURE 1 Eight sections, in order, with about how long each takes.

**Time. About 35 minutes for the lesson, then about 15 for the five exercises.**

# 14.1 weighted_average, Docstring First

THE BOOK'S DURATION, AGAINST EXCEL

```
importlib.reload(bondmath)
book_duration_file = bondmath.weighted_average(list(holdings['duration']),
                                               list(holdings['market_value']))
excel_duration = 6.254777411342787
print(f'weighted_average {book_duration_file:.15f}, Excel {excel_duration:.15f}')
```

```
weighted_average 6.254777411342787, Excel 6.254777411342787
```

EXCEL SIDE BY SIDE

| JOB | PYTHON | EXCEL | RESULT |
|---|---|---|---|
| Book value | `holdings['market_value'].sum()` | `=SUMPRODUCT(R2:R201,I2:I201+K2:K201)/100` | 490,819,215.00 |
| Book duration | `bondmath.weighted_average(...)` | `=SUMPRODUCT(N2:N201,S2:S201)/SUM(S2:S201)` | 6.254777411342787 |

`weighted_average` **returns Excel's SUMPRODUCT over SUM to the last digit: the book's duration is 6.254777 years.**

# 14.2 The Book's Risk Card

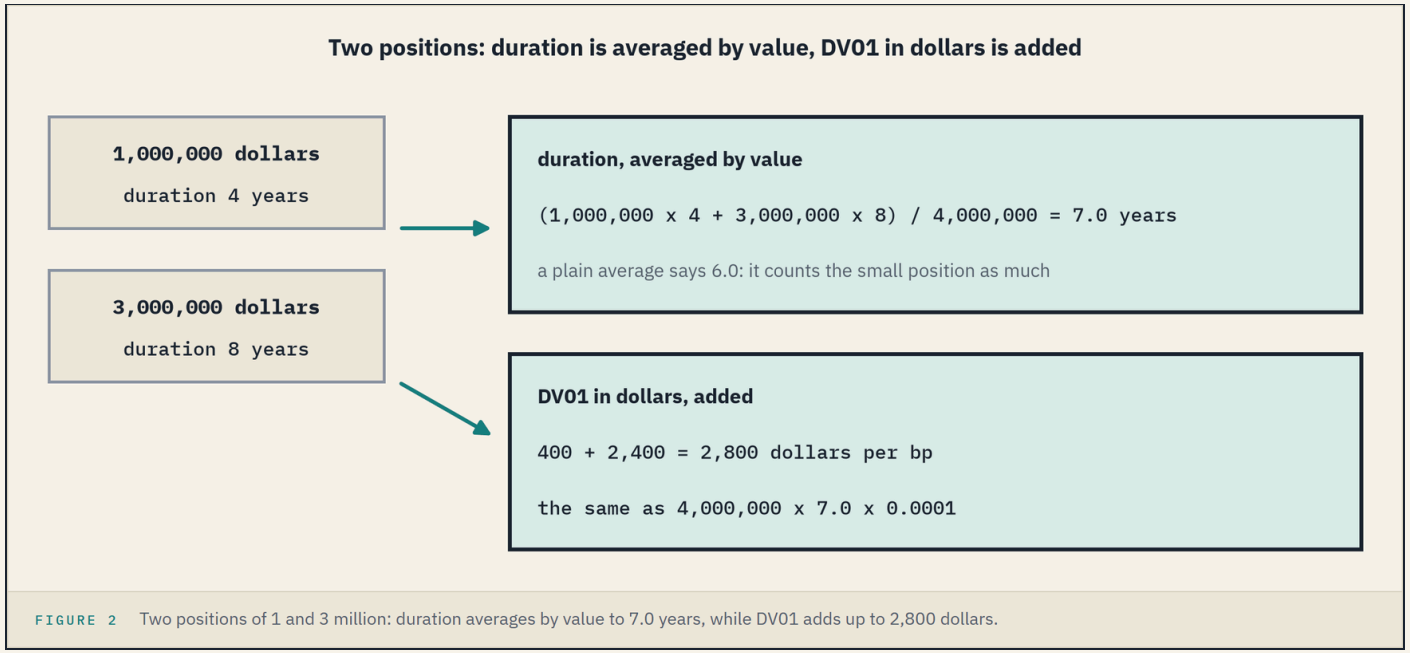



FIGURE 2 Two positions of 1 and 3 million: duration averages by value to 7.0 years, while DV01 adds up to 2,800 dollars.

THE BOOK IN SIX NUMBERS

| | |
|---|---|
| market value | 490,819,215.00 dollars |
| duration | 6.2548 years |
| spread | 180.46 bp |
| DTS | 1064.11 years x bp |
| convexity | 66.9912 years squared |
| DV01 | 306,996.11 dollars per bp |

**Four of the six are averages weighted by value and DV01 is a sum: this card is the base every what-if moves from.**

# 14.3 Every Yield Moves, Repriced in Full

BEZANE, 100 BP HIGHER

```
settlement, maturity, coupon_pct, yield_pct = bonds[0]
before = bondmath.dirty_price(settlement, maturity, coupon_pct, yield_pct)
after = bondmath.dirty_price(settlement, maturity, coupon_pct, yield_pct + 100 / 100)
change_dollars = holdings.loc[0, 'par_held'] * (after - before) / 100
print(f'dirty {before:.4f} at {yield_pct} percent, {after:.4f} at {yield_pct + 1:.3f}')
print(f'the position changes by {change_dollars:,.2f} dollars')
```

```
dirty 108.7099 at 5.606 percent, 103.8202 at 6.606
the position changes by -244,484.16 dollars
```

THE BASE THE CHANGES ARE MEASURED FROM

```
base_value = (holdings['par_held'] * holdings['dirty_bm'] / 100).sum()
print(f'book from the yields  {base_value:,.2f} dollars')
print(f"book from the file    {total_value:,.2f} dollars, {total_value - base_value:,.2f} more")
```

```
book from the yields  490,819,038.14 dollars
book from the file    490,819,215.00 dollars, 176.86 more
```

# 14.4 Every Spread Moves

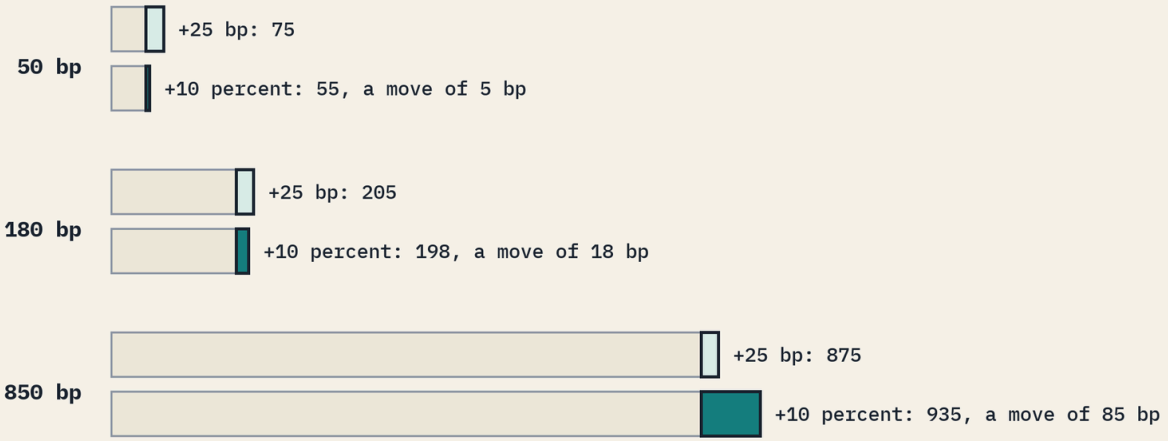



FIGURE 4 Spreads of 50, 180 and 850 bp: +25 bp moves each by 25, +10 percent by 5, 18 and 85.

BOOK DTS GIVES THE 10 PERCENT MOVE

```
from book DTS   -1.0641 percent
full reprice    -1.0493 percent
```

**The book's spread is 180.46 bp, so its average move is about 18 bp: less than 25, and so a smaller loss.**

# 14.5 price_change_estimate, Against the Full Reprice

**price_change_estimate: Bezane Finance, 100 bp higher, per 100 of face**

```
dirty_price x (-modified_duration x dy + 0.5 x convexity x dy ** 2)
```

| dirty_price | modified_duration | convexity | dy |
|---|---|---|---|
| 108.7099 | 4.626971 | 26.3580 | 100 / 10000 = 0.01 |
| per 100 of face | years | years squared | shift_bp as a decimal |




FIGURE 5 Bezane at 100 bp: the duration term of -5.0300 and convexity term of +0.1433 give -4.8867.

BEZANE, 100 BP

| | |
|---|---|
| full reprice | -4.8897 per 100 |
| duration and convexity | -4.8867 per 100 |
| duration alone | -5.0300 per 100 |

**Duration alone misses by 0.14 per 100; the convexity term takes the miss to 0.003.**

# The Shortcut Against the Book

THE ESTIMATE, IN THE MODULE

```
def price_change_estimate(dirty_price: float, modified_duration: float, convexity: float, shift_bp: float) -> float:
    dy = shift_bp / 10000
    return dirty_price * (-modified_duration * dy + 0.5 * convexity * dy ** 2)
```

THE BOOK AT 25 BP

```
for shift_bp in [-25, 25]:
    full = totals[f'rates {shift_bp:+d} bp']
    miss = full - estimate([shift_bp] * len(holdings)).sum()
    print(f'{shift_bp:+d} bp: full {full:,.2f}, the shortcut misses by {miss:,.2f} dollars')
```

```
-25 bp: full 7,778,969.05, the shortcut misses by 1,314.62 dollars
+25 bp: full -7,573,434.14, the shortcut misses by -1,283.05 dollars
```

# 14.6 Where Each Loss Lands, by Sector

DIVIDED BACK

| SECTOR | WEIGHT_PCT | DURATION | DTS | DV01_SHARE_PCT |
|---|---|---|---|---|
| Materials | 14.61 | 5.26 | 863.68 | 12.29 |
| Financials | 13.90 | 6.22 | 1006.25 | 13.82 |
| Consumer | 11.54 | 6.97 | 1395.78 | 12.86 |
| Telecom | 11.38 | 6.16 | 1090.25 | 11.20 |
| Industrials | 10.70 | 6.70 | 1567.20 | 11.46 |
| Transportation | 9.95 | 8.19 | 914.17 | 13.02 |
| Health Care | 7.93 | 5.13 | 637.25 | 6.50 |
| Energy | 7.91 | 6.06 | 1025.17 | 7.67 |
| Technology | 6.18 | 5.47 | 869.87 | 5.40 |
| Utilities | 5.91 | 6.12 | 1166.57 | 5.78 |

EXCEL SIDE BY SIDE

| JOB | EXCEL | RESULT |
|---|---|---|
| Transportation's duration | the calculated field's cell | 8.186803162389122 |
| The whole book | Grand Total | 6.254777411342787 |

# 14.7 The Grid to the Desk, the Code to Git

THE WORKBOOK

```
with pd.ExcelWriter('scenario_grid.xlsx') as writer:
    grid.round(2).to_excel(writer, sheet_name='scenarios')
    sector_pnl.round(2).to_excel(writer, sheet_name='by_sector')
```

READ IT BACK

```
back = pd.read_excel('scenario_grid.xlsx', sheet_name=None, index_col=0)
assert np.isclose(back['scenarios'], grid.round(2)).all()
assert np.isclose(back['by_sector'], sector_pnl.round(2)).all()
print(list(back.keys()), 'read back, every number as written')
```

```
['scenarios', 'by_sector'] read back, every number as written
```

WHAT GIT SEES

```
 M bondmath.py
  M test_bondmath.py
?? scenario_grid.xlsx
```

STAGED BY HABIT

```
M  bondmath.py
A  scenario_grid.xlsx
M  test_bondmath.py
```

**The desk gets the workbook; the record gets the 59 lines that make it.**

# Bring It Home

COPY AND TEST

```
Set-Location "$HOME\projects\my_bondmath"
$work = "$env:TEMP\pfi_course2_14_k4w7q2xr"
Copy-Item "$work\bondmath.py", "$work\test_bondmath.py", "$work\.gitignore" .
python -m pytest -q
```

CHECK AND COMMIT

```
git diff --stat
git add bondmath.py test_bondmath.py
git status --short
```

**If a file slipped in, `git restore --staged <file>` takes it out before the commit.**

# You Can Now

- Weight a book by market value with **`weighted_average`**, as Excel's **SUMPRODUCT** over **SUM** does.
- Build a book's risk card from your own functions, and say which numbers are averages and which add up.
- Reprice a whole book in full under parallel shifts and under two shapes of spread move, and say what a scenario is and is not.
- Estimate the same moves with **`price_change_estimate`**, and say when the shortcut stops being enough.
- Split each scenario's loss by sector, with **`groupby`** or a PivotTable, and tie the parts back to the book.
- Send a scenario grid as a workbook, and take a file out of the staging area with **`git restore --staged`**.

NEXT

WEEK 3 · DAY 5

## Course 2 Capstone

November's extract proved with your own bondmath, cleaned by one function, reconciled, reported, rerun on September, and written up. You write the code.